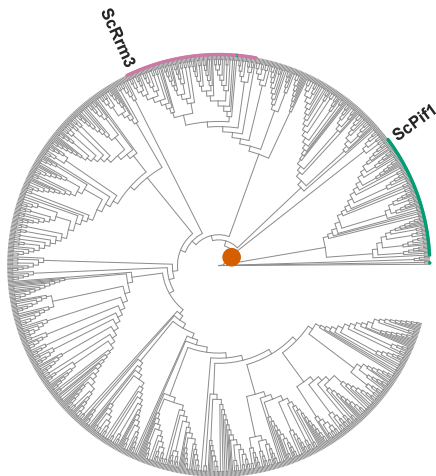


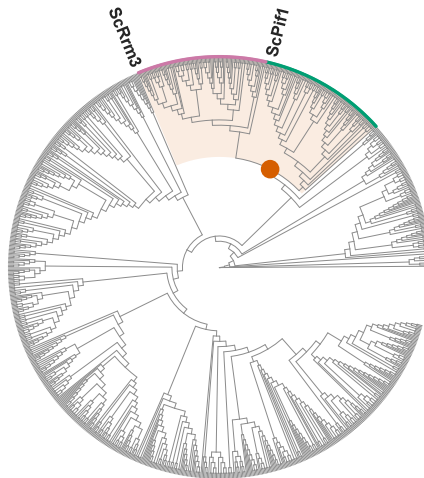
amino acids alone

Pif1/Rrm3 MRCA -> 950 of 957 tips (21% Saccharomycotina)



amino acids + 3Di structure

Pif1/Rrm3 MRCA -> 197 of 957 tips (100% Saccharomycotina)



● Pif1 clade ● Rrm3 clade ● Pif1/Rrm3 duplication

grams of 957 PIF1-family proteins; tips coloured by paralog (Pif1 vs Rrm3 daughter clade). Adding the 3Di structural alphabet collapses the Pif1/Rrm3 duplication from the base of Fungi to